



PythonX Bootcamp

Python Foundations for Data Science

Assignment 02

Total Marks: 20

Instructions

- Answer all questions.
- Write **Python code with proper comments**.
- Use **clear variable names** and **good programming practices**.
- Submit your work as a **.py file or Jupyter/Colab Notebook (.ipynb)**.

Question 1: You are a junior data scientist at a healthcare startup. Your team needs to analyze patient vitals collected over 7 days. The data is stored as raw Python lists and must be converted into NumPy arrays for efficient processing. **(5 Marks)**

Write a Python program using NumPy that: (1) creates a 1D NumPy array of 7 patient heart rate readings with realistic values, (2) reshapes it into a **(7, 1)** column array and prints the result along with its shape, (3) computes and displays the mean, max, and min heart rate using NumPy functions, and (4) creates a second array of 7 temperature readings and adds it element-wise to a normalized version of the heart rate array using broadcasting.

Question 2: A university research lab is tracking exam scores of 4 students across 3 subjects. The data is arranged in a 2D matrix and you need to extract and compare performance using array slicing. **(5 Marks)**

Write a Python program using NumPy that: (1) creates a **(4, 3)** NumPy array representing scores of 4 students in 3 subjects, (2) slices and prints the scores of the first two students only, (3) extracts and displays the second subject's scores for all students using column indexing, and (4) uses boolean indexing to print only the scores that are above 70.

Question 3: Your AI team is building an image classification model. Before feeding data into the model, you must understand how tensors differ from NumPy arrays and how to prepare them correctly using TensorFlow. **(5 Marks)**

Write a Python program using TensorFlow that: (1) creates a `tf.constant` tensor from a 3×3 list of values and prints its `.shape` and `.dtype`, (2) reshapes the tensor to shape `(1, 9)` using `tf.reshape()` and prints the result, (3) applies `tf.expand_dims()` to add a new axis and `tf.transpose()` to swap dimensions and prints both outputs, and (4) creates an equivalent NumPy array and prints a side-by-side comparison of the shape, dtype, and type of both objects.

Question 4: You are working on a medical imaging project where X-ray images need to be pre-processed and your supervisor has asked you to visualize an image, analyze its pixel intensity distribution, and compare the original and grayscale versions side by side. **(5 Marks)**

Write a Python program using Matplotlib that: (1) loads any sample image using `matplotlib.image.imread()` or a sample from `skimage` and displays it using `plt.imshow()` with a proper title, (2) converts the image to grayscale and plots a pixel intensity histogram using `plt.hist()` with labeled axes, (3) uses `plt.subplot(1, 2, ...)` to display the original color image and grayscale image side by side with titles for each, and (4) adds a main title using `plt.suptitle()` and saves the final figure as a `.png` file.